Unit 2
Physical Layer

Introduction

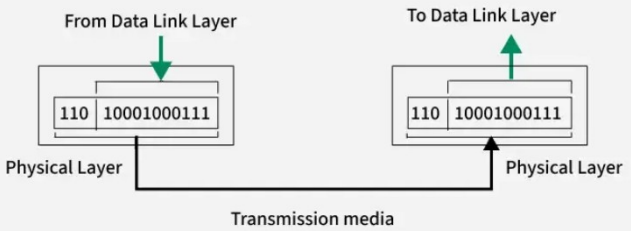
- The physical Layer is the bottom-most layer in the Open System Interconnection (OSI) Model, responsible for the physical and electrical transmission of data. It consists of various network components such as power plugs, connectors, receivers, cable types, etc. The physical layer sends data bits from one device(s) (like a computer) to another device(s).

- Functions of Physical Layer
  - The Physical Layer is responsible for sending raw data as bits over a physical medium.
  - It converts data into signals that can travel through wires, fiber optics, or wireless channels (encoding) and turns these signals back into data at the receiver (decoding).
  - It ensures signals are transmitted correctly and uses techniques like modulation to prepare the data for transmission and demodulation to retrieve it at the other end.
  - This layer also decides how data flows (one-way, two-way alternately, or simultaneously) through transmission modes and controls the speed and timing of data transmission to keep everything running smoothly.

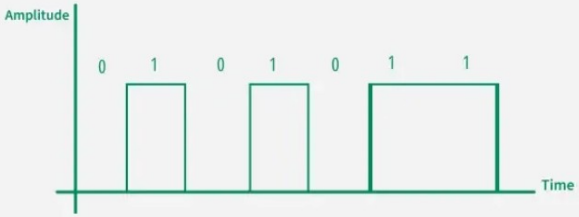
Functions of Physical Layer

1. Bit-by-Bit Transmission
2. Encoding and Decoding
3. Signal Transmission
4. Modulation and Demodulation
5. Transmission Modes
6. Data Control

1. Bit-by-Bit Transmission



2. Encoding and Decoding

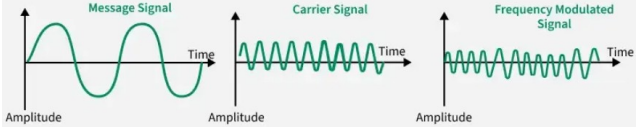


3. Signal Transmission

It is responsible for converting data into signals (analog or digital) for transmission and decoding these signals at the receiver's end.

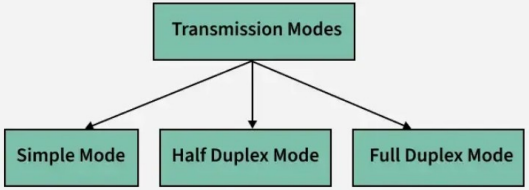
4. Modulation and Demodulation

Modulation is the process of modifying a carrier signal's properties (amplitude, frequency, or phase) for transmission over a communication medium.



5. Transmission Modes

Physical Layer determines the direction of data flow.



6. Data Rate Control

Physical Layer controls the speed of data transmission, ensuring the sender and receiver can handle the same data rate.

Physical Topologies

- Line Configuration:
- Point-to-Point configuration: In Point-to-Point configuration, there is a line (link) that is fully dedicated to carrying the data between two devices.
- Multi-Point configuration: In a Multi-Point configuration, there is a line (link) through which multiple devices are connected.

Transmission Media in Computer Networks

- 1. Guided Media
  - Guided Media is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links. There are 3 major types of Guided Media: Twisted Pair, Coaxial and Optical Fiber Cables
  - Features:
    - High Speed
    - Secure
    - Used for comparatively shorter distances

- 1.1 Twisted Pair Cable
  - It consists of 2 separately insulated conductor wires twisted about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media. Twisted Pair is of two types:
  - (a) Unshielded Twisted Pair (UTP): UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Advantages:

- Least expensive in comparison
- Easy to install
- High-speed capacity

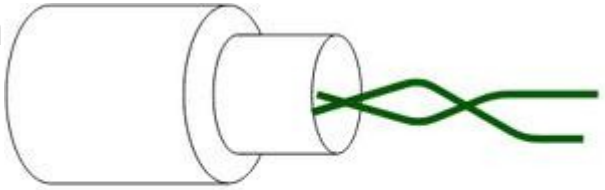
Unshielded Twisted Pair

Disadvantages:

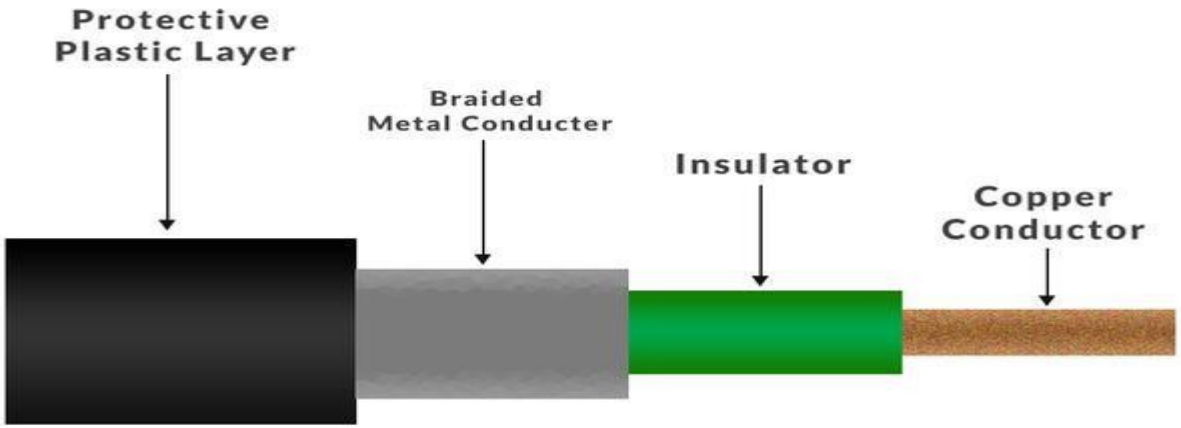
Lower capacity and performance to STP.

Short distance transmission due to attenuation

- (b) Shielded Twisted Pair (STP): Shielded Twisted Pair (STP) cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference. It is used in fast data rate Ethernet and in voice and data channels of telephone lines.
- Advantages:
  - Better performance at a higher data rate in comparison
  - Eliminates crosstalk
  - Comparatively faster
- Disadvantages:
  - Comparatively difficult to install and manufacture
  - More expensive
  - Bulky



Shielded Twisted Pair

- 1.2 Coaxial Cable:
  - Coaxial cable has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover. The coaxial cable transmits information in two modes: Baseband mode(dedicated cable bandwidth) and Broadband mode(cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.
- 

- Advantages:
  - Coaxial cables has high bandwidth .
  - It is easy to install.
  - Coaxial cables are more reliable and durable.
  - Less affected by noise or cross-talk or electromagnetic inference.
  - Coaxial cables support multiple channels
- Disadvantages:
  - Coaxial cables are expensive.
  - The coaxial cable must be grounded in order to prevent any crosstalk.
  - As a Coaxial cable has multiple layers it is very bulky.
  - There is a chance of breaking the coaxial cable and attaching a “t-joint” by hackers, this compromises the security of the data.

- 1.3 Optical Fiber Cable
  - Optical Fibre Cable uses the concept of total internal reflection of light through a core made up of glass. The core is surrounded by a less dense glass or plastic covering called the coating. It is used for the transmission of large volumes of data. The cable can be unidirectional or bidirectional. The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.

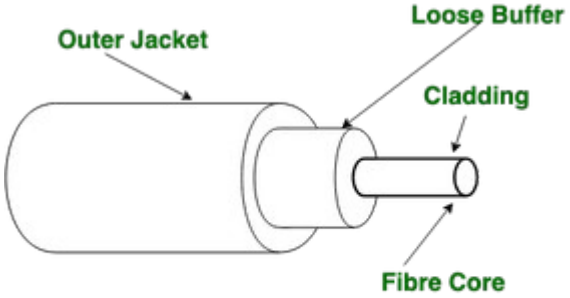
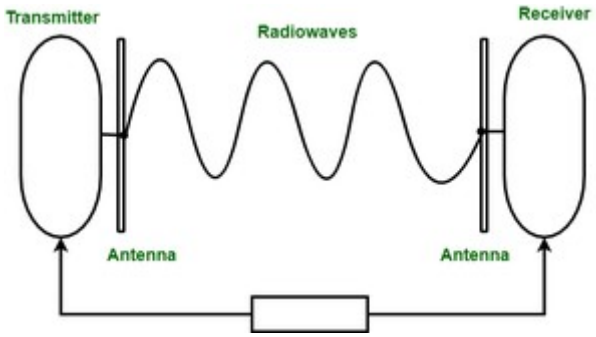


Figure of Optical Fibre Cable

- Advantages:
  - Increased capacity and bandwidth
  - Lightweight
  - Less signal attenuation
  - Immunity to electromagnetic interference
  - Resistance to corrosive materials
- Disadvantages:
  - Difficult to install and maintain
  - High cost
- Applications:
  - Medical Purpose: Used in several types of medical instruments.
  - Defence Purpose: Used in transmission of data in aerospace.
  - For Communication: This is largely used in formation of internet cables.

- 2. Wireless Transmission(Unguided Media):
  - It is also referred to as Wireless or Unbounded transmission media . No physical medium is required for the transmission of electromagnetic signals. There are 3 types of Signals transmitted through unguided media: Radio wave, Microwave and Infrared Wave.
- Features:
  - The signal is broadcasted through air
  - Less Secure
  - Used for larger distances

- 2.1 Radio Waves:
  - Radio waves are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range:
  - 3KHz - 1GHz
  - 3KHz - 1GHz. AM and FM radios and cordless phones use Radio waves for transmission.
- Types of Radio Waves:
  - Short Wave: AM Radio
  - VHF (Very High Frequency): FM Radio/TV
  - UHF (Ultra High Frequency): TV
- Radio Wave Components:
    - Transmitter: Responsible for encoding the signal.
    - Receiver: Responsible for decoding the signal.



- 2.2 Microwaves:
  - It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range:1GHz - 300GHz. Micro waves are majorly used for mobile phone communication and television distribution.
- Advantages:
  - Cheaper than using cables
  - Freedom from land acquisition
  - Ease of communication in difficult terrains
  - Communication over oceans

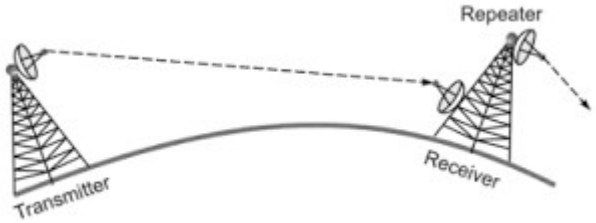
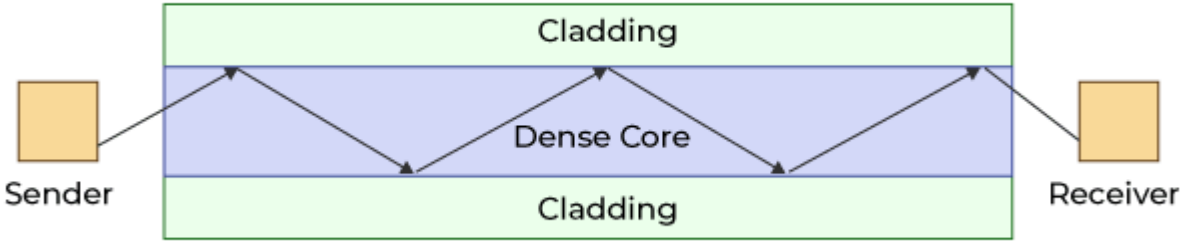


Fig: Microwave Transmission

- Disadvantages:
  - Insecure communication.
  - Out of phase signal.
  - Susceptible to weather conditions.
  - Bandwidth is limited.
  - High cost of design, implementation, and maintenance.

- 2.3 Infrared:
  - Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems. Frequency Range:300GHz - 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.
  -

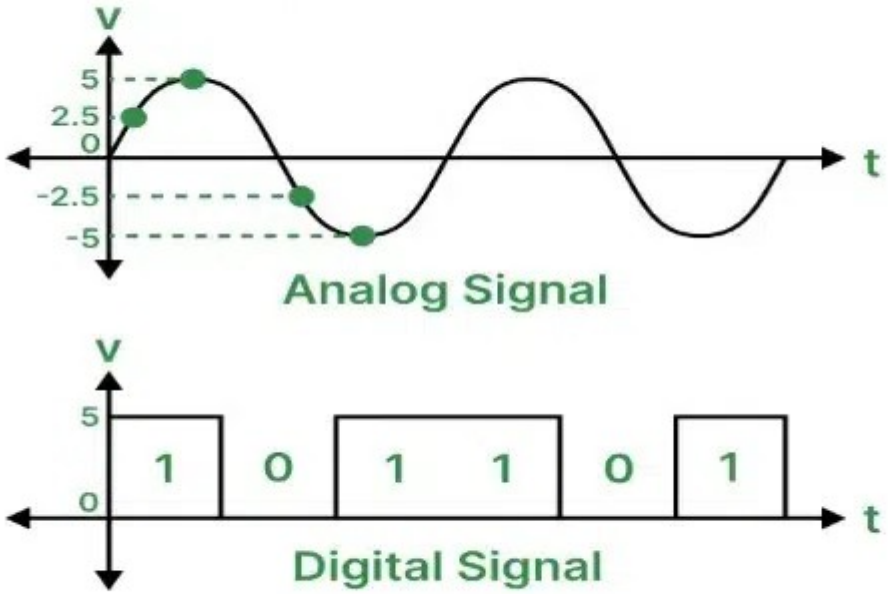
- What is Optical Fiber Light Transmission?
  - Optical Fiber Light Transmission commonly known as fiber optics is a technology that utilizes thin transparent fibers made of glass or plastic to transmit data and information using the light signals. It is a method of transmitting data and video over long distances through the propagation of light along these optical fibers.



- What is Modulation?
  - Modulation is the process of superimposing high-frequency carrier signals with low-frequency message signals resulting in a modulated wave. In modulation, one of the three parameters of the carrier wave is changed i.e. amplitude, frequency or phase according to the amplitude of the message signal at a given time instant. It is an important technique used to increase the range of communication, increase the signal-to-noise ratio, and decrease the size of the antenna.
  - There are mainly two types of modulation techniques which are
  - Analog Modulation
  - Digital Modulation

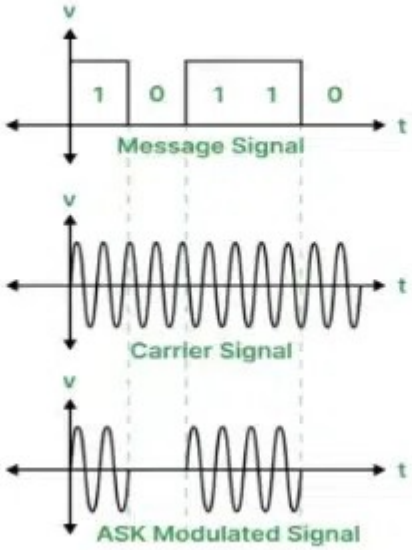
- Analog Modulation : In Analog Modulation, both carrier and message signals are analogue waves.
- Digital Modulation : In Digital Modulation, only the carrier signal is analogue and the message signal is in digital form.
- Analog waves : Analog waves are those signals which can take an infinite value between two points.
- Digital signals : Digital signals are binary values that can take either logic 0 as in zero volts or logic 1 as in any other voltage. There are no in-between voltage levels in the digital signal.

Types of Signals

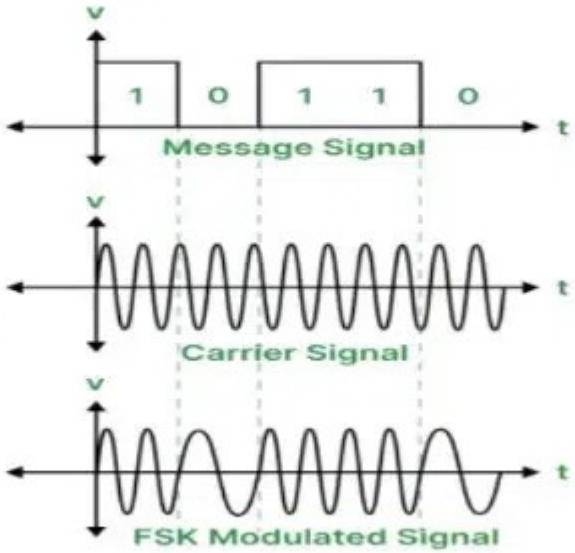


- What are Digital Modulation Techniques ?
  - There are mainly three types of Digital Modulation techniques:
    - Amplitude Shift Keying
    - Frequency Shift Keying
    - Phase Shift Keying

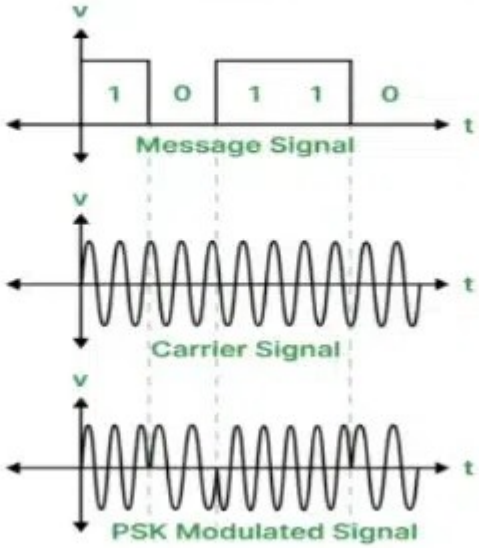
Amplitude Shift Keying Waveform



Frequency Shift Keying Waveform

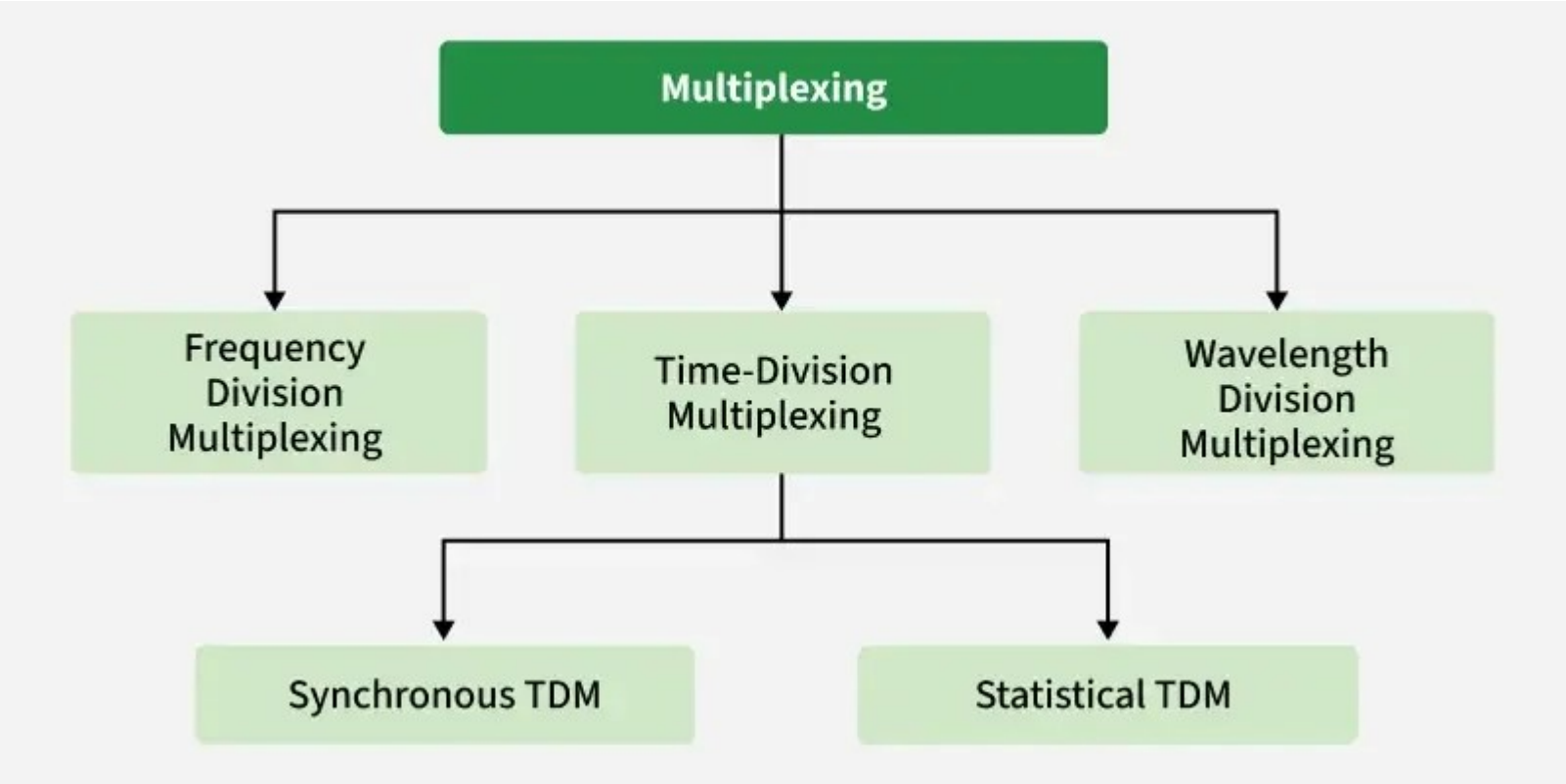


Phase Shift Keying Waveform

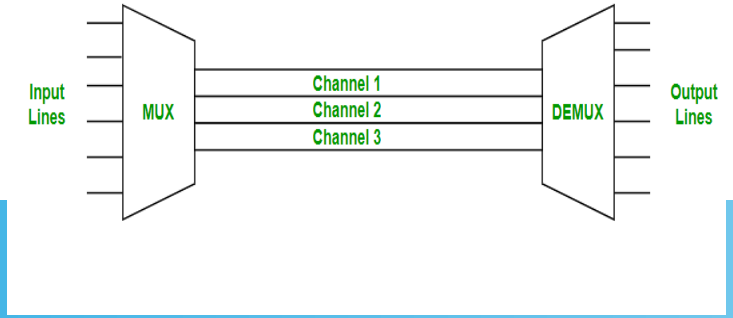


Digital Modulation Techniques

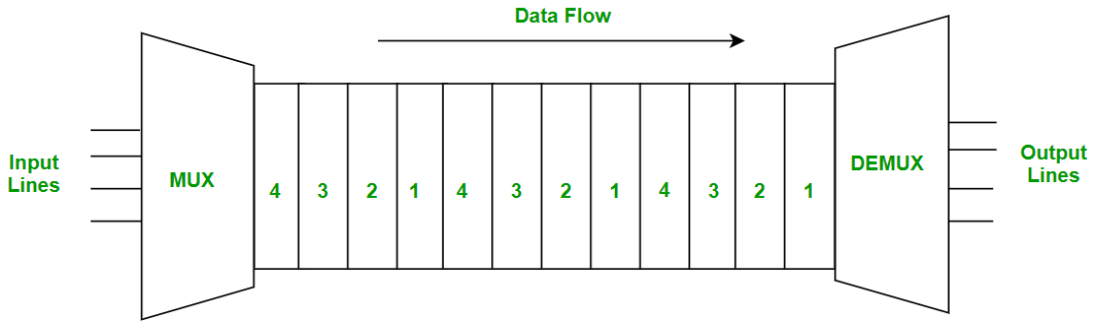
Types of Multiplexing



- 1. Frequency Division Multiplexing
  - Frequency division multiplexing is defined as a type of multiplexing where the bandwidth of a single physical medium is divided into a number of smaller, independent frequency channels.
    - In FDM, we can observe a lot of inter-channel cross-talk because in this type of multiplexing the bandwidth is divided into frequency channels.
    - In order to prevent the inter-channel cross talk, unused strips of bandwidth must be placed between each channel. These unused strips between each channel are known as guard bands.



- 2. Time Division Multiplexing
  - Time-division multiplexing is multiplexing wherein FDM, instead of sharing a portion of the bandwidth in the form of channels, in TDM, time is shared. Each connection occupies a portion of time in the link. In Time Division Multiplexing, all signals operate with the same frequency (bandwidth) at different times.



- 3. Wavelength Division Multiplexing
  - Wavelength Division Multiplexing (WDM) is a multiplexing technology used to increase the capacity of optical fiber by transmitting multiple optical signals simultaneously over a single optical fiber, each with a different wavelength.
  - Each signal is carried on a different wavelength of light, and the resulting signals are combined onto a single optical fiber for transmission.
  - At the receiving end, the signals are separated by their wavelengths, demultiplexed and routed to their respective destinations.
  - It is used in telecommunications, cable TV, ISPs, and data centers for high-speed, long-distance data transmission.

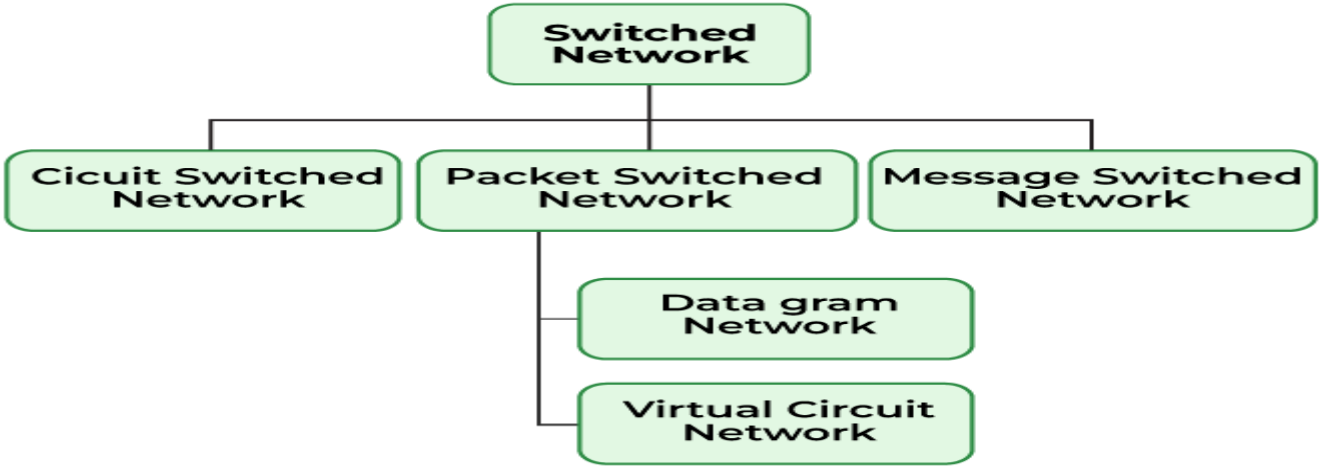
- Advantages of Multiplexing
  - Efficient Use of Bandwidth: You can send more than one signal over a single channel. This way, you can use the channel's capacity more efficiently.
  - Increased Data Transmission: Multiplexing can significantly boost the amount of data that can be sent over a network simultaneously, enhancing overall transmission capacity.
  - Scalability: Multiplexing allows networks to easily expand and accommodate more data streams without requiring significant changes to the existing infrastructure.
  - Flexibility: Different types of multiplexing (TDM, FDM, WDM, CDM) can be used based on the specific needs and characteristics of the communication system, providing flexibility in network design.

- Disadvantages of Multiplexing
  - Synchronization Issues: Ensuring that multiple data streams remain properly synchronized can be challenging, leading to potential data loss or errors if not managed correctly.
  - Latency: Combining multiple signals into one can introduce delays, as each data stream needs to be processed, synchronized, and demultiplexed at the receiving end.
  - Signal Degradation: Over long distances, multiplexed signals can experience degradation and interference, requiring additional measures such as signal boosters or repeaters to maintain quality.
  - Resource Management: Allocating and managing resources for multiplexing can be complicated, requiring careful planning and real-time adjustments to avoid congestion and ensure efficient operation.

Switching

- Switching is the process of transferring data packets from one device to another within the same network or across networks using switches. It happens at the Data Link Layer (Layer 2) of the OSI Model, right after the Network Layer generates data packets.
  - A switch is a device that connects multiple devices in a network.
  - It ensures data from different devices does not interfere.
  - Works like a traffic controller by sending data packets to the correct port.
  - Sends data directly to connected devices (e.g., computer, phone).
  - For other networks, it forwards data to a router for delivery.
  - Example: When you open a website, your request is broken into packets and forwarded through various switches before reaching the server.

- Types of Switching:
  - There are three types of switching methods:
    - Message Switching
    - Circuit Switching
    - Packet Switching



- Message Switching: This is an older switching technique that has become obsolete. In message switching technique, the entire data block/message is forwarded across the entire network thus, making it highly inefficient.
- Circuit Switching: In this type of switching, a dedicated connection is established between sender and receiver before communication starts. This connection receives the complete bandwidth of the network until the data is transferred completely.
- Packet Switching: In packet switching, data is divided into small packets that travel independently through the network using available paths. Each packet carries its destination address, so switches and routers know where to send it. At the destination, packets are reassembled into the original message. This is the method used in modern networks and the Internet because it is fast and efficient.